EE558 – Optical Fiber Comm. Systems
Prof. Alan E. Willner

Homework #1

For the questions below, use the following constants:

Speed of light: $c = 2.998 \times 10^{8}$ m/s
Electron charge: $e = 1.602 \times 10^{-9}$ C
Plank constant: $h = 6.626 \times 10^{-34}$ J/Hz

1- (a) Visible light has wavelengths that range roughly from 0.4 to 0.7 μm. What is the corresponding range of frequencies in Hertz?
(b) If a bit corresponds to a single, square pulse that is present for a "1" and absent for a "0":
i) What is the maximum duration of each pulse if the transmission rate is 1 Mbit/sec? 1 Gbit/sec?
ii) For each rate in i), how much energy is carried by a "1" pulse if its power level is 1 mW?
(c) If window glass has losses equivalent to 10^{5} dB/km, what fraction of the original light energy remains after it traverses 5 mm of such glass?
(d) What is the speed of light in glass of refractive index n=1.5?

2- A fiber path 50 km long consists of fiber, each having a maximum length of 3 km, spliced together to form one continuous fiber. The attenuation of the fiber is 0.5 dB/km; the splices have additional loss of 0.3 dB per splice. Connectors at each end, to connect the fiber to the transmitter and the receiver, have added loss of 1 dB per connector. (a) What is the total loss, in dB, between the transmitter and the receiver? (b) What is the power input to the receiver when the power output to the transmitter is 1 mW?

3- (a) The indices of refraction for the materials in a slab waveguide are n_1=1.49 and n_2=1.48. The wavelength is 1.3 μm and the waveguide radius is 100 μm. Calculate the number of modes in this waveguide assuming we can use the formula given in the class.
(b) Plot the number of modes as a function of the wavelength of light (Remember modes are discrete).

4- Envision the interface between two regions, one glass (n_g=1.5) and the other of water (n_w=1.33). A ray traveling in the glass impinges on the interface at 45 degrees and refracts into the water. What is the transmission angle?

5- Determine the critical angle for a water (n_w=1.33) and glass (n_g=1.5) interface. Does it matter from which medium the wave is coming from?

6- (a) What is the oscillating frequency of a 1.55μm optical signal?
(b) An optical filter with 0.1 nm bandwidth is centered at the optical signal (i.e., at 1.55μm).

What is the bandwidth of the optical filter in Hertz?

7- (a) What is the energy (unit: eV) of the photon in vacuum at 0.82 μm, 1.3 μm and 1.55 μm, respectively?
(b) If the photon has energy of 1.24 eV, what is the emission wavelength? What is the optical frequency of the emitted light when propagating along fiber? (Assume the refractive index of the fiber is 1.475)

8- In a Fabry-Perot cavity, the cavity is 430 μm long and the refractive index is set to 3.6.
(a) How many transmission resonances exist between 1.53 μm and 1.57 μm? Assume the light is normally incident on the cavity.
(b) If the cavity is 0.43 μm long, how many modes exist between 1.53 μm and 1.57 μm?

9- As illustrated in the figure below, one consider a step index fiber for which n_1=1.475, n_2=1.460 and the radius of the core r=25 μm. Ray A and Ray B carry information at the same bit rate, and θ is the maximum angle for which the rays will be guided through the fiber. They will reach to the end of the fiber with a time delay and thus case "inter-modal dispersion".
(a) Derive the expression of the pulse spreading. (Hint: express the time delay between Ray A and Ray B as a function of the parameters given in the figure.)
(b) Calculate the pulse spreading per kilometer.
(c) The number of spatial modes can be expressed as $\frac{2r}{\lambda}\sqrt{n_1^2 - n_2^2}$. Calculate the number of spatial modes if the wavelength of the light is set to 1.55 μm.